Calculus I

Lecture #4

Outline

1. Derivatives of Functions
2. Differentiation Rules
3. The Chain Rule
4. Implicit Differentiation
5. Related Rates
6. Linear Approximations

We have the following differentiation formulas which allow us to differentiate functions without having to take limits each time.

$$\frac{d}{dx}(c) = 0 \qquad \frac{d}{dx}(x^s) = sx^{s-1}$$

$$(cf)' = cf' \quad \text{Constant multiple} \qquad (f \pm g)' = f' \pm g' \quad \text{Sum/Diff.}$$

$$(fg)' = f'g + fg' \quad \text{Product} \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad \text{Quotient}$$

Comments. Please avoid the following common mistakes.

$$(fg)' \neq f'g', \quad \left(\frac{f}{g}\right)' \neq \frac{f'}{g'}, \quad \left(\frac{f}{g}\right)' \neq \frac{fg' - gf'}{g^2}.$$

- Derivatives of trigonometric functions

The differentiation formulas for trigonometric functions are in the following table. Remember that they are valid only when x is measured in radians.

$$(\sin x)' = \cos x \qquad (\csc x)' = -\csc x \cot x$$

$$(\cos x)' = -\sin x \qquad (\sec x)' = \sec x \tan x$$

$$(\tan x)' = \frac{1}{\cos^2 x} = \sec^2 x \qquad (\cot x)' = -\frac{1}{\sin^2 x} = -\csc^2 x$$

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There is a question: How to differentiate a composite function $f \circ g$?

Theorem (The Chain Rule)

If $g(x)$ is differentiable at x and $f(u)$ is differentiable at $u = g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$ are differentiable, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where $\dfrac{dy}{du}$ is evaluated at $u = g(x)$.

To use the Chain Rule, we apply the so-called "outer-inner" principle:

- First, find and differentiate the outer function.
- Next, evaluate what you get at the point $g(x)$.
- Finally, multiply that by the derivative of the innner function, evaluated at x.

Example

Apply the chain rule to differentiate

$$F(x) = \sin \frac{x^2 - 1}{x^2 + 1}.$$

Solution. This function is a composition $F(x) = f \circ g(x)$, with:

- the outer function: $f(x) = \sin x$,
- the inner function: $g(x) = \dfrac{x^2 - 1}{x^2 + 1}$.

Note that both of $f(x)$ and $g(x)$ are differentiable at every point of $\mathbb{R}$. Then we can use the Chain Rule to evaluate the derivative of $F(x)$ at an arbitrary point x.

- **Step 1**. Differentiate the outer function $f(x) = \sin x$:

$$f'(x) = \cos x.$$

- **Step 2**. Evaluate the result of Step 1 at $g(x)$ (the value of the inner function at the point x). This is

$$\cos \frac{x^2 - 1}{x^2 + 1}.$$

- **Step 3**. Multiply the result of Step 2 by the derivative of the inner function:

$$\begin{aligned}
\frac{dF}{dx} &= \cos \frac{x^2 - 1}{x^2 + 1} \cdot \frac{d}{dx}\left[\frac{x^2 - 1}{x^2 + 1}\right] \\
&= \cos \frac{x^2 - 1}{x^2 + 1} \cdot \frac{(2x)(x^2 + 1) - (2x)(x^2 - 1)}{(x^2 + 1)^2} \\
&= \cos \frac{x^2 - 1}{x^2 + 1} \cdot \frac{4x}{(x^2 + 1)^2}.
\end{aligned}$$

• The Power Rule combined with the Chain Rule

Let us make explicit the special case of the Chain Rule where the outer function f is a power function.

If $y = [g(x)]^s$, then we can write $y = f(u) = u^s$ where $u = g(x)$. By using the Chain Rule and then the Power Rule, we get

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = su^{s-1}\frac{du}{dx} = s\,[g(x)]^{s-1}\,g'(x).$$

So we obtain the following result.

Theorem

If s is a real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^s) = su^{s-1}\frac{du}{dx} \quad \Longleftrightarrow \quad \frac{d}{dx}[g(x)]^s = s\,[g(x)]^{s-1}\,g'(x).$$

Example

Differentiate

(a) $y = \sin(x^2)$

(b) $y = \sin^2 x$.

Solution.

(a) For $y = \sin(x^2)$, we have the outer function is $\sin u$, while the inner function is x^2. So the Chain Rule gives

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot (2x) = 2x \cos(x^2).$$

(b) For $y = \sin^2 x = (\sin x)^2$, we have the outer function is u^2, while the inner function is $\sin x$. So the Chain Rule gives

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 2u \cdot \cos x = 2 \sin x \cos x = \sin 2x.$$

The reason for the name Chain Rule becomes clear when we make a longer chain by adding another link.

Suppose that $y = f(u)$, $u = g(x)$, and $x = h(t)$, where f, g, and h are differentiable functions. Then, to compute the derivative of y with respect to t, we use the Chain Rule twice:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{dy}{du} \cdot \frac{du}{dx} \cdot \frac{dx}{dt}.$$

It is similar for more functions.

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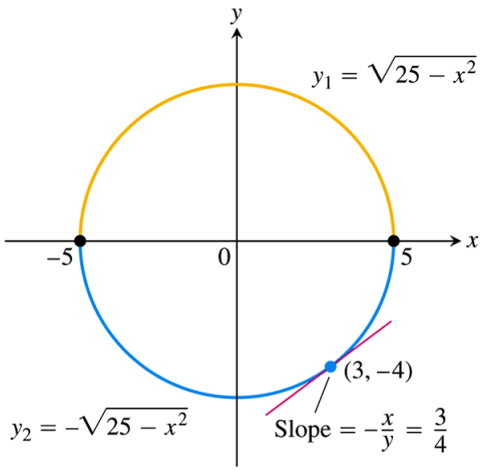
• Implicit functions

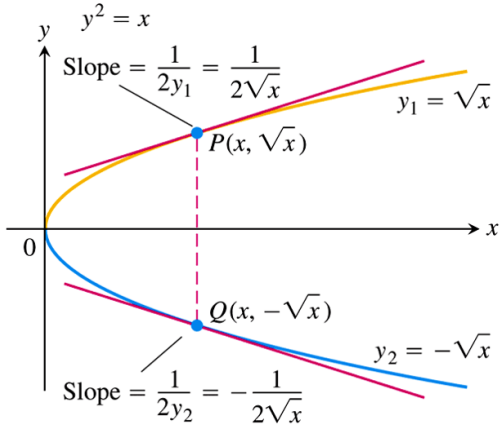
The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable, $y = f(x)$.

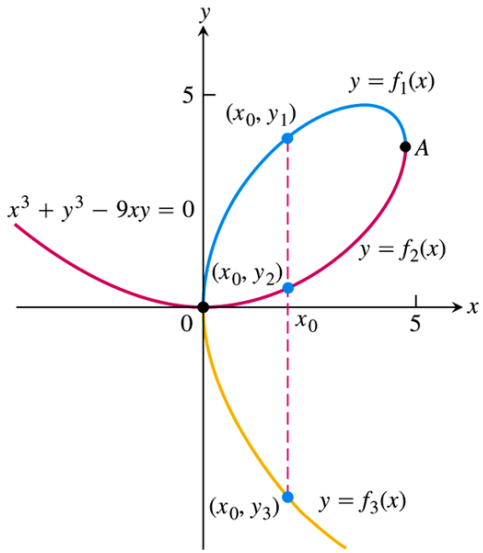
However, some functions are defined implicitly by a relation between x and y, say $F(x, y) = 0$, such as

$$x^2 + y^2 = 25, \quad y^2 - x = 0, \quad x^3 + y^3 - 9xy = 0.$$

The curves represented these equations are as follows.


y
x
−5
0
5
$y_1 = \sqrt{25 - x^2}$
$y_2 = -\sqrt{25 - x^2}$
(3, −4)
Slope $= -\frac{x}{y} = \frac{3}{4}$


$y^2 = x$
y
x
0
Slope $= \frac{1}{2y_1} = \frac{1}{2\sqrt{x}}$
$y_1 = \sqrt{x}$
$P(x, \sqrt{x})$
$Q(x, -\sqrt{x})$
$y_2 = -\sqrt{x}$
Slope $= \frac{1}{2y_2} = -\frac{1}{2\sqrt{x}}$


y
x
5
0
x_0
5
$y = f_1(x)$
(x_0, y_1)
A
$x^3 + y^3 - 9xy = 0$
$y = f_2(x)$
(x_0, y_2)
(x_0, y_3)
$y = f_3(x)$

In some cases, it is possible to solve such an equation for y as an explicit function (or even several functions) of x.

- First, the circle $x^2 + y^2 = 25$ combines the graphs of two functions $y_1(x) = \sqrt{25 - x^2}$ and $y_2(x) = -\sqrt{25 - x^2}$, which are respectively the upper (yellow) and lower (blue) semicircles.
- Next, the equation $y^2 - x = 0$ defines into two differentiable functions $y_1(x) = \sqrt{x}$ (yellow) and $y_2(x) = -\sqrt{x}$ (blue).
- Finally, the curve $x^3 + y^3 - 9xy = 0$ is not the graph of any one function of x, but it can be divided into separate arcs which are the graphs of functions of x (yellow, blue, and red).

• Implicit Differentiation

When we cannot solve an equation for y in terms of x in order to find the derivative of y, fortunately, we may still be able to find dy/dx by the technique of implicit differentiation.

Implicit differentiation.

- Differentiate both sides of the equation w.r.t x (considering y as a differentiable function of x)
- Collect the terms with $\dfrac{dy}{dx}$ on one side of the equation and solve for $\dfrac{dy}{dx}$.

In the following example, it is (always) assumed that y is a differentiable function of x.

Example

(a) If $x^2 + y^2 = 25$, find $\dfrac{dy}{dx}$.

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$

Solution.

(a) - Differentiate both sides of the equation:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25) \quad \Longleftrightarrow \quad \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0.$$

Remembering that y is a function of x and using the Chain Rule, we have

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2)\frac{dy}{dx} = 2y\frac{dy}{dx}.$$

Thus

$$2x + 2y\frac{dy}{dx} = 0.$$

- Now we solve this equation for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{x}{y}.$$

(b) Equation of the tangent at the point $(3, 4)$

We have $x = 3$ and $y = 4$, and so

$$\frac{dy}{dx} = -\frac{3}{4}.$$

Then the equation of the tangent to the circle at $(3, 4)$ is

$$y - 4 = -\frac{3}{4}(x - 3) \quad \Longleftrightarrow \quad 3x + 4y = 25.$$

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• Related Rates

In the real life, there are many problems that ask for the rate at which some variable changes when it is known how the rate of some other related variables changes.

Such a problem of finding a rate of change from other known rates of change is called a related rates problem.

For example, if we pump air into a balloon, both the volume and the radius of the balloon increase, and their rates of increase are highly related to each other.

But it is much easier to measure the increasing rate of the volume than the increasing rate of the radius.

Thus the essential idea of related rate problem is to compute the rate of change of one quantity, in terms of the rate of change of another quantity (which may be more easily measured).

The procedure

- First, find an equation of the two quantities
- Next, use the Chain Rule (to differentiate both sides w.r.t. time) to get the relation between their rates.

First we look at the following simple example.

Example (Tom and Jerry)



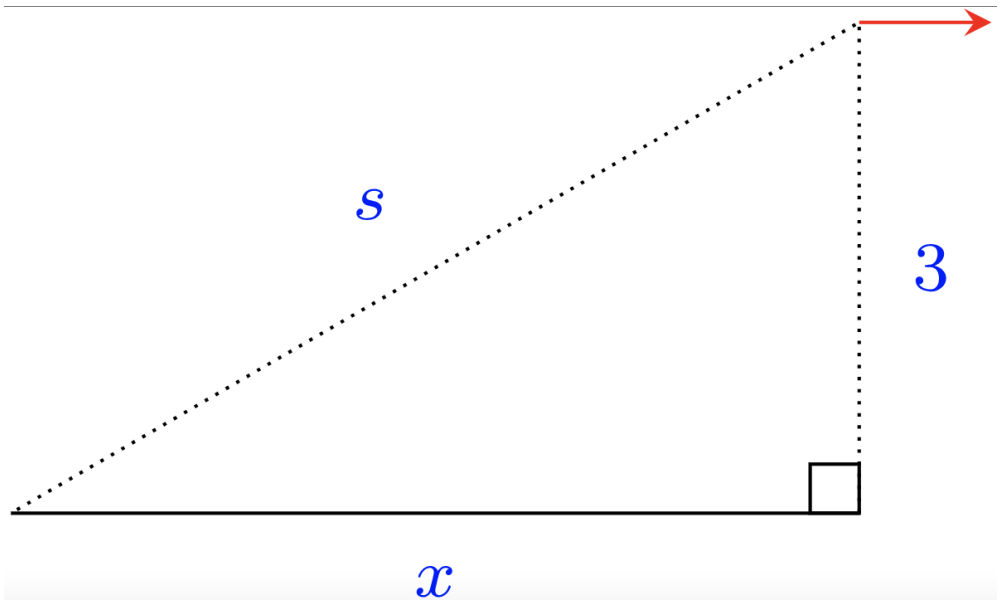




Example (Tom and Jerry - cont'd)

A Jerry's plane is flying directly away from Tom at 500 miles per hour at an altitude of 3 miles. How fast is the Jerry's plane distance from Tom increasing at the moment when the Jerry's plane is flying over a point on the ground 4 miles from Tom?

Solution. We first sketch a schematic representation of the situation.



We are dealing with rates of changing w.r.t. to a time t.

From the assumptions of the problem, we note the following:

- because the Jerry's plane is in level flight directly away from Tom, the rate at which x changes is the speed of the plane, i.e. $\frac{dx}{dt} = 500$ mph.
- the distance between Tom and the Jerry's plane is s.
- it is $\frac{ds}{dt}$ we wish to know.

By the Pythagorean Theorem, $x^2 + 3^2 = s^2$. Taking the derivative with respect to the independent variable t, we obtain

$$2x\frac{dx}{dt} = 2s\frac{ds}{dt}.$$

We are interested in the time at which $x = 4$, knowing that at this time

$$4^2 + 9 = s^2 \implies s = 5.$$

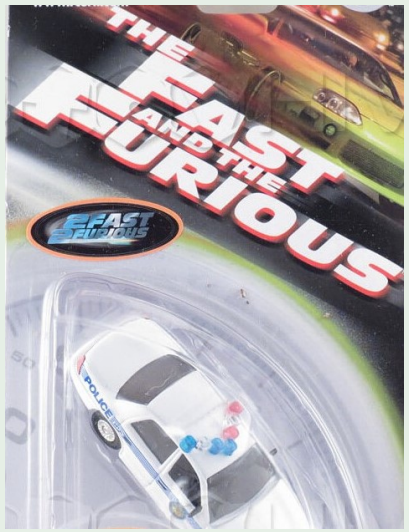
Therefore,

$$2 * 4 * 500 = 2 * 5 * \frac{ds}{dt},$$

which gives

$$\frac{ds}{dt} = 400 \text{ miles per hour}.$$

Example (The Fast & the Furious)





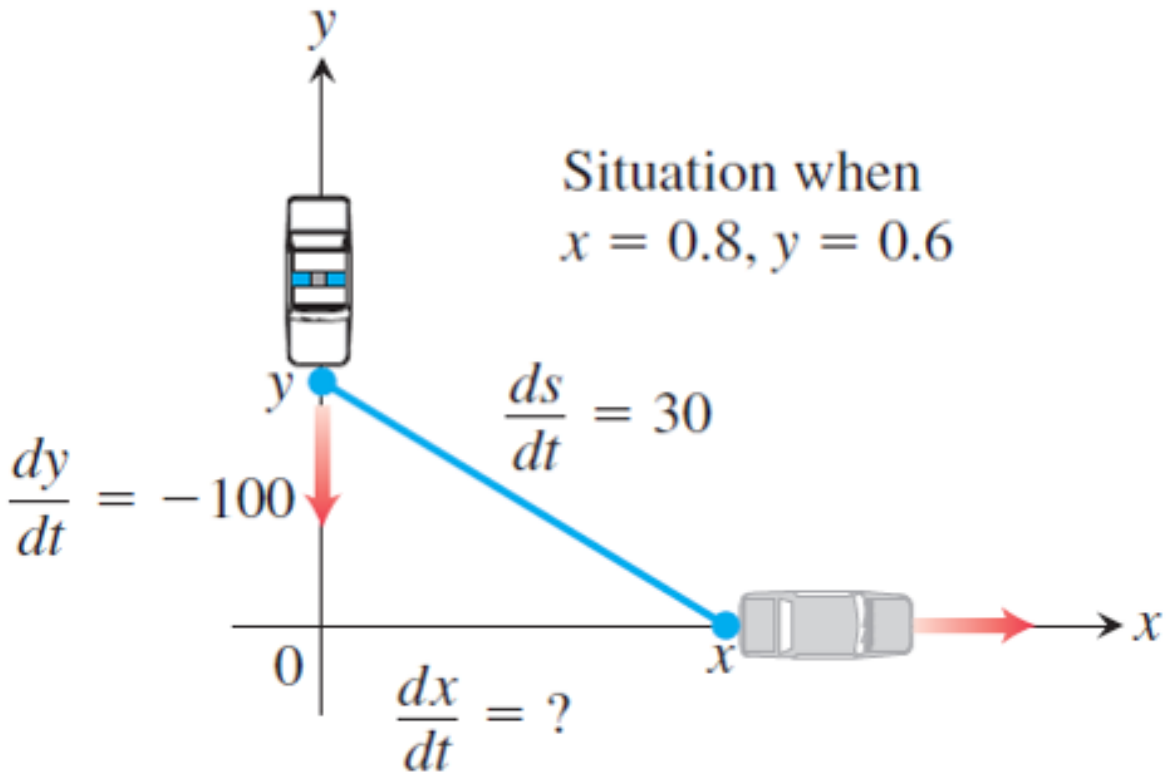


Example (The Fast & the Furious - cont'd)

A police car, approaching the right-angled intersection from the North, is chasing a speeding car (driven by Dominic Torreto) that has turned the corner and is now moving straight East. When the police car is 0.6 km North of the intersection and Dom's car is 0.8 km to the East, the police officer determines with radar that the distance between two cars is increasing at 30 km/h. Suppose the police car is moving at 100 km/h, what is the speed of Dom's car?

Solution. We use the coordinate plane to draw a diagram.

Let x, y, and s are respectively a position of speeding car, a position of police car, and a distance between two cars, all at time t. Also we assume x, y, and s are all differentiable functions w.r.t t.


y
Situation when
$x = 0.8, y = 0.6$
y
$\frac{ds}{dt} = 30$
$\frac{dy}{dt} = -100$
0
x
x
$\frac{dx}{dt} = ?$

The given info is $x = 0.8$ km, $y = 0.6$ km, $\frac{dy}{dt} = -100$ km/h, $\frac{ds}{dt} = 30$ km/h and the unknown is $\frac{dx}{dt}$.

Note that $\frac{dy}{dt} < 0$, because y is decreasing.

We have

$$s^2 = x^2 + y^2,$$

and hence we differentiate it to get

$$2s\frac{ds}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt} \iff \frac{ds}{dt} = \frac{1}{\sqrt{x^2 + y^2}}\left(x\frac{dx}{dt} + y\frac{dy}{dt}\right).$$

Substituting $x = 0.8$, $y = 0.6$, $\frac{dy}{dt} = -100$, $\frac{ds}{dt} = 30$ into this equation, we obtain

$$30 = \frac{1}{\sqrt{(0.8)^2 + (0.6)^2}} \left(0.8 \frac{dx}{dt} + 0.6(-100) \right)$$

$$\Updownarrow$$

$$\frac{dx}{dt} = \frac{30\sqrt{(0.8)^2 + (0.6)^2} + 0.6(100)}{0.8} = 112.5$$

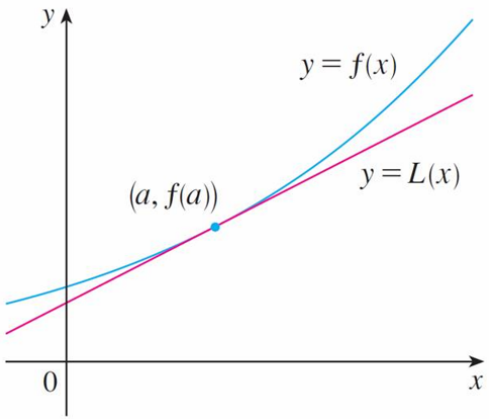
Thus at the moment in the question, the speeding car's speed is 112.5 km/h.

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• Linearization

Let us consider the Figure below.



Near the point of tangency, the curve always lies close to its tangent line. In fact, by zooming in toward a point on the graph, we see that the graph looks more and more like its tangent line.

This observation provides a way to approximate values of functions, called a linearization

The idea is that it might be easy to calculate the value of function $f(x)$ at a point $x = a$, but difficult (or even impossible) to compute $f(x)$ at nearby points ($x \approx a$).

So we approximate the $f(x)$ at ($x \approx a$) by the linear function $L(x)$, whose graph is the tangent line of $f(x)$ at $(a, f(a))$.

In other words, we use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a.

The equation of this tangent line is

$$y = f(a) + f'(a)(x - a).$$

Definitions. Let f be a differentiable function at a point $x = a$. The approximating function

$$L(x) = f(a) + f'(a)(x - a)$$

is the linearization of f at a.

- The approximation

$$f(x) \approx L(x)$$

  of f by L is called the standard linear approximation of f at a.
- The point $x = a$ is called the the center of the approximation.

Example

Find a linear approximation of function $f(x) = \sqrt{x+3}$ at $a = 1$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$. Are these approximations overestimates or underestimates?

Solution. The derivative of $f(x) = \sqrt{x+3}$ is

$$f'(x) = \frac{1}{2\sqrt{x+3}} \quad \Longrightarrow \quad f(1) = 2 \quad \text{and} \quad f'(1) = \frac{1}{4}.$$

The linear approximation of function $f(x)$ at $x = 1$ is

$$L(x) = f(1) + f'(1)(x-1) = 2 + \frac{1}{4}(x-1) = \frac{7}{4} + \frac{x}{4}.$$

Hence, the corresponding linear approximation of f at $x = 1$ is

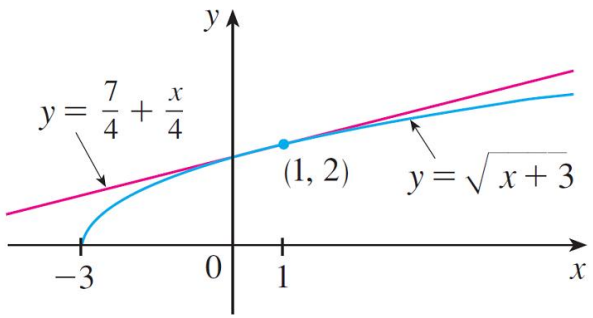
$$\sqrt{x+3} \approx \frac{7}{4} + \frac{x}{4}.$$

Taking, in particular, $x = 0.98$ and $x = 1.05$, we have

$$\sqrt{3.98} \approx \frac{7}{4} + \frac{0.98}{4} = 1.995,$$

and

$$\sqrt{4.05} \approx \frac{7}{4} + \frac{1.05}{4} = 2.0125.$$



The two approximations are all overestimates, because the tangent line lies above the curve.

• Differentials

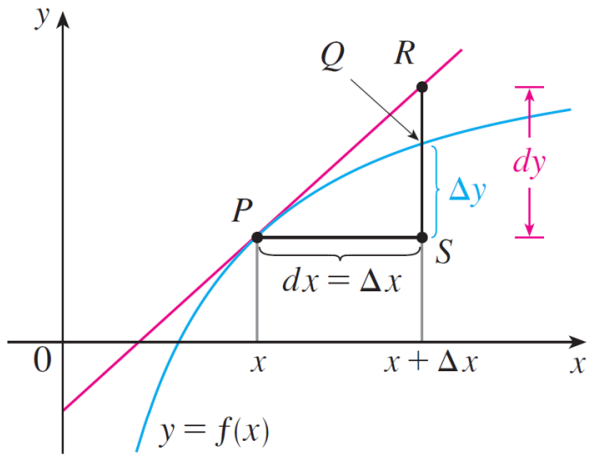
The ideas behind linear approximations can be better understood through differentials.

For a differentiable function $y = f(x)$, we have $y' = f'(x) = \dfrac{dy}{dx}$. But contrary to its appearance, it is not a ratio.

We now introduce two new variables dx and dy, with the properties that when their ratio exists, it is equal to the derivative.

Definitions. Let $y = f(x)$ be a differentiable function. The differential dx is an independent variable. The differential dy is

$$dy = f'(x)\, dx.$$



Geometrically, if for a point $P(x, f(x))$ on the graph of $f(x)$, we increase the coordinate x by a small value Δx, the change in x, then we arrive at a new point $Q(x + \Delta x, f(x + \Delta x))$ on the graph.

Often the variable dx is chosen to be Δx, ; i.e. $dx = \Delta x$. Then the change in y is $\Delta y = f(x + \Delta x) - f(x)$.

Since the differential

$$dy = f'(x)\,dx,$$

we see that *dy* represents the amount that the tangent line rises or falls (the change in the linearization), when x changes by an amount dx.

On the other hand, Δy represents the amount that the curve $y = f(x)$ rises or falls, when x changes by an amount $dx = \Delta x$. That is,

$$\Delta y = f(x + \Delta x) - f(x).$$

Example

Compare the values of Δy and dy if $y = f(x) = x^3 + x^2 - 2x + 1$ and x changes

(a) from 2 to 2.05

(b) from 2 to 2.01.

Solution. (a) We have $\begin{cases} f(2) = f(x)\Big|_{x=2} = 9, \\ f(2.05) = f(x)\Big|_{x=2.05} = 9.717625. \end{cases}$.

Then $\Delta y = f(2.05) - f(2.0) = 0.717625$.

On the other hand,

$$dy = f'(x)dx = (3x^2 + 2x - 2)\, dx,$$

and hence when $x = 2$ and $dx = \Delta x = 0.05$, this becomes

$$dy = (3x^2 + 2x - 2)\Big|_{x=2}(0.05) = 0.7.$$

(b) We have $f(2.01) = f(x)\Big|_{x=2.01} = 9.140701$, and hence

$$\Delta y = f(2.01) - f(2.0) = 0.140701.$$

When $x = 2$ and $dx = \Delta x = 0.01$, this becomes

$$dy = (3x^2 + 2x - 2)\Big|_{x=2}(0.01) = 0.14.$$

Comment. From the example, we can see that the differential dy can be used as an approximation of Δy. Clearly, compute dy is easier than finding Δy.

Example

The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

Solution. Let r be the radius of the sphere. Then the volume V of the sphere is $V = \frac{4}{3}\pi r^3$.

If the error of r (in the measured value) is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV, which can be approximated by the differential $dV = V'dr = 4\pi r^2\, dr$.

When $r = 21$ and $dr = 0.05$, this becomes

$$dV = 4\pi(21)^2 \cdot (0.05) \approx 277.$$

So the maximum error in the calculated volume is about 277 cm^3.

END OF CHAPTER 2